

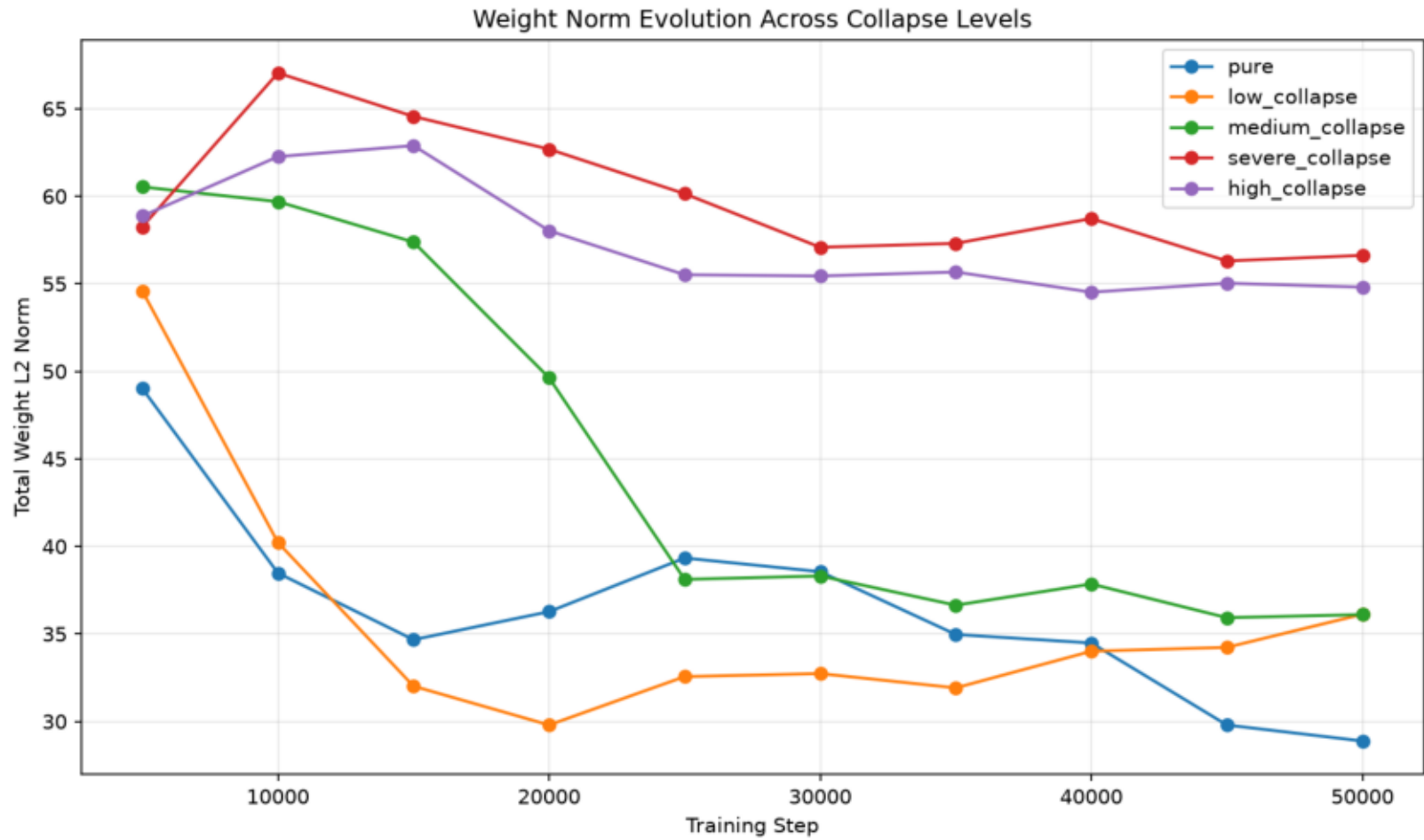
Grokking & Model Collapse: Mechanistic Interpretability Report

Generated Analysis Suite

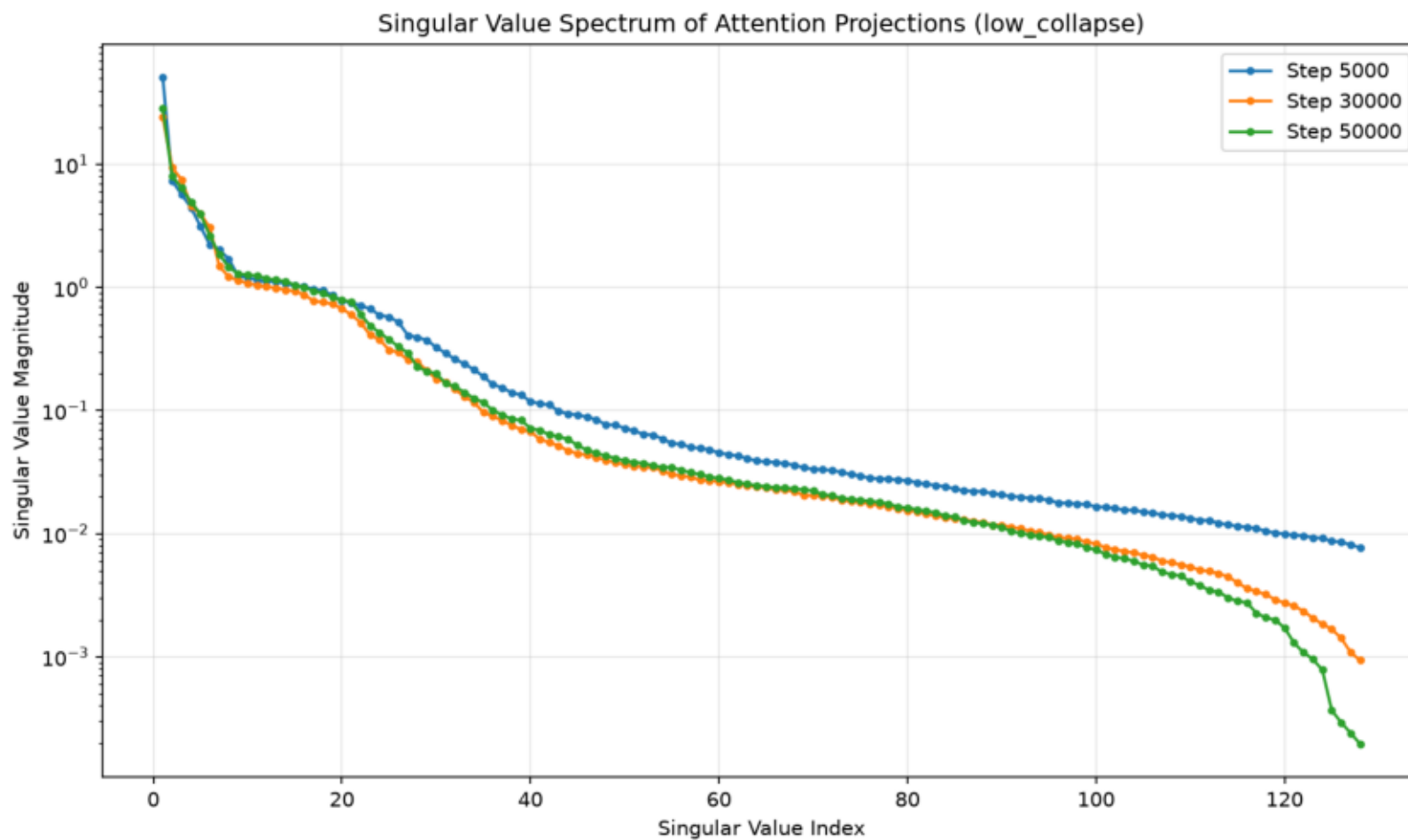
Circuit Formation Summary:

```
high_collapse: Grok=-1, Rank Drop=5000  
low_collapse: Grok=3100, Rank Drop=5000  
medium_collapse: Grok=-1, Rank Drop=5000  
  pure: Grok=1400, Rank Drop=5000  
severe_collapse: Grok=-1, Rank Drop=5000
```

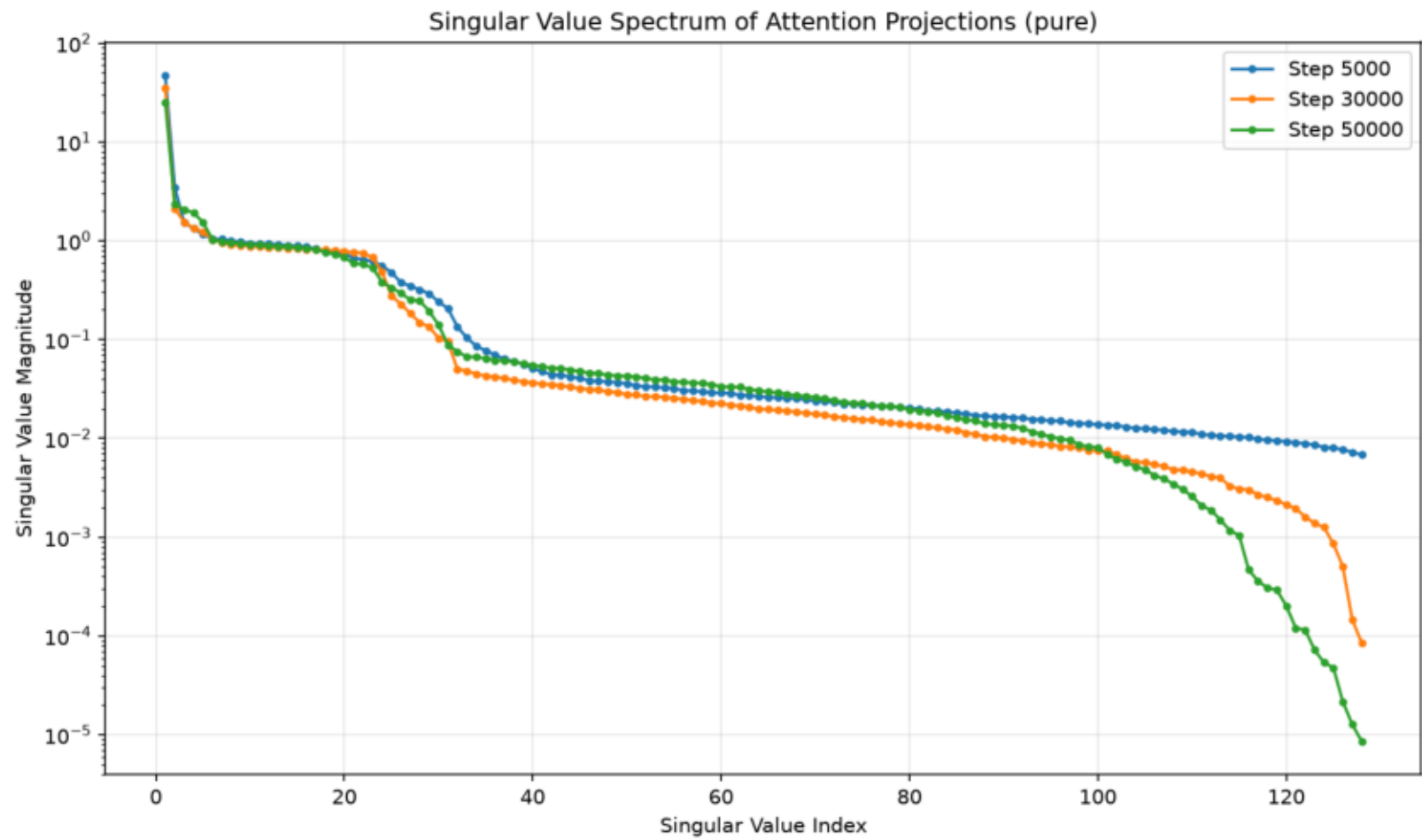
Weight Norm Evolutions



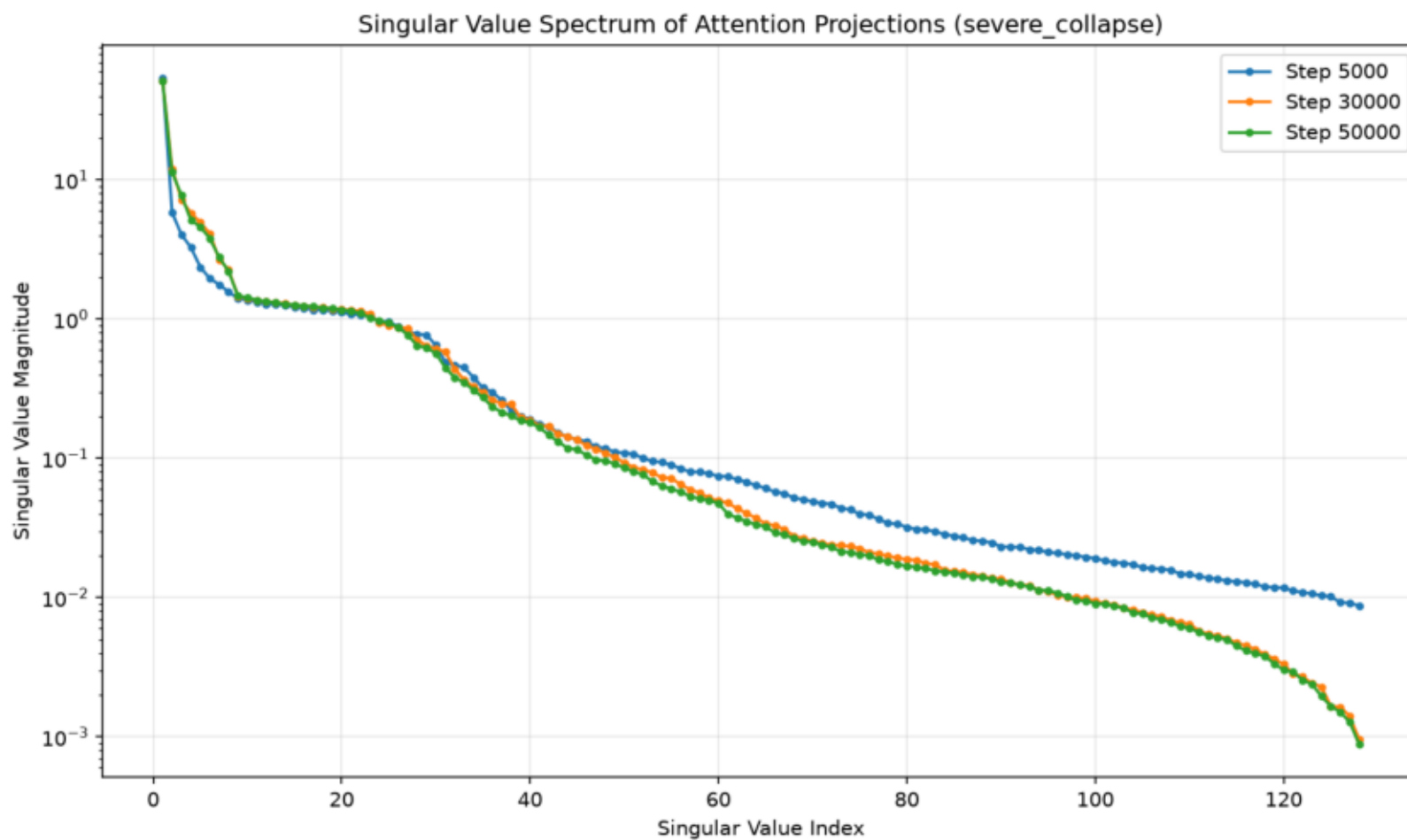
Singular Value Spectrum (low)



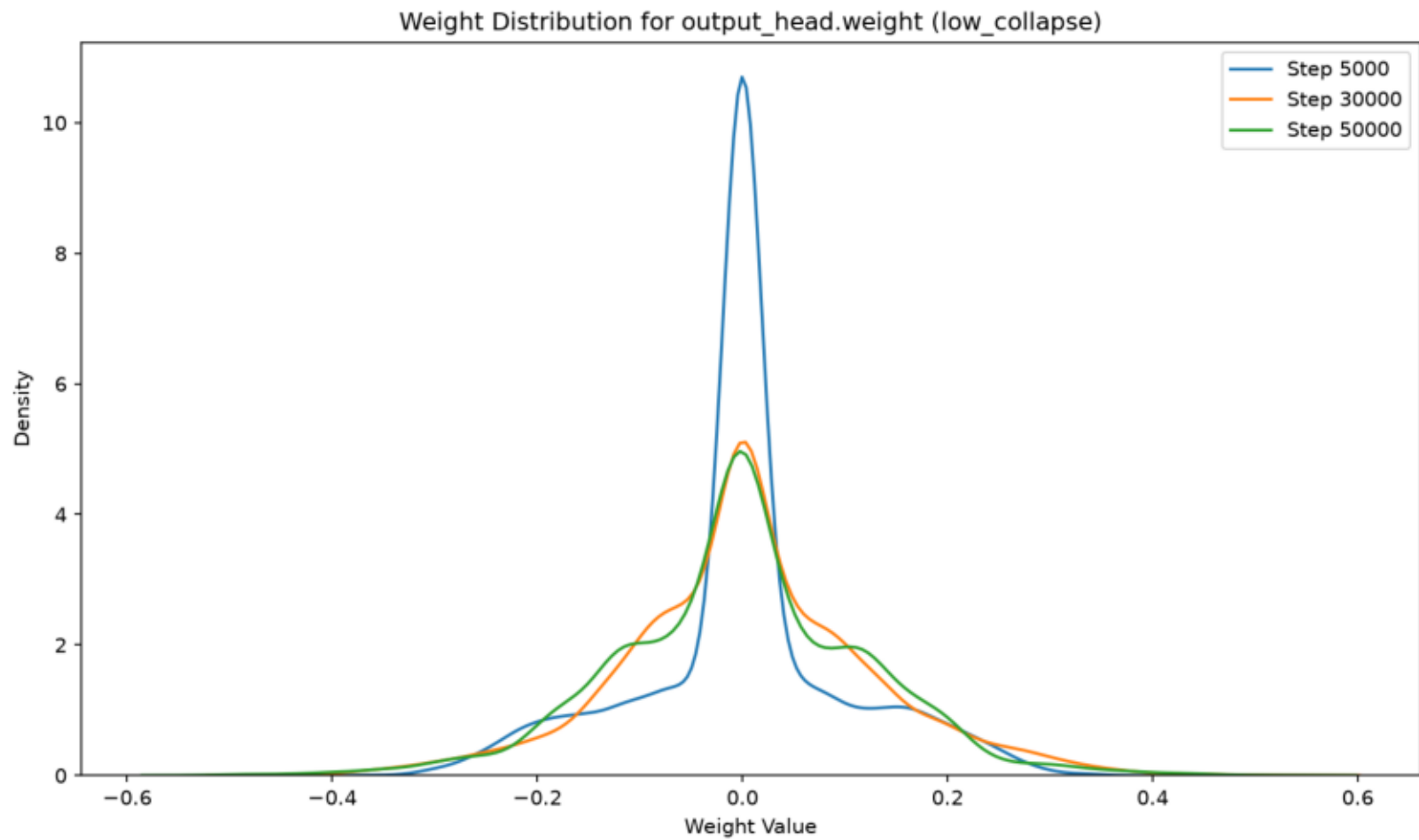
Singular Value Spectrum (pure)



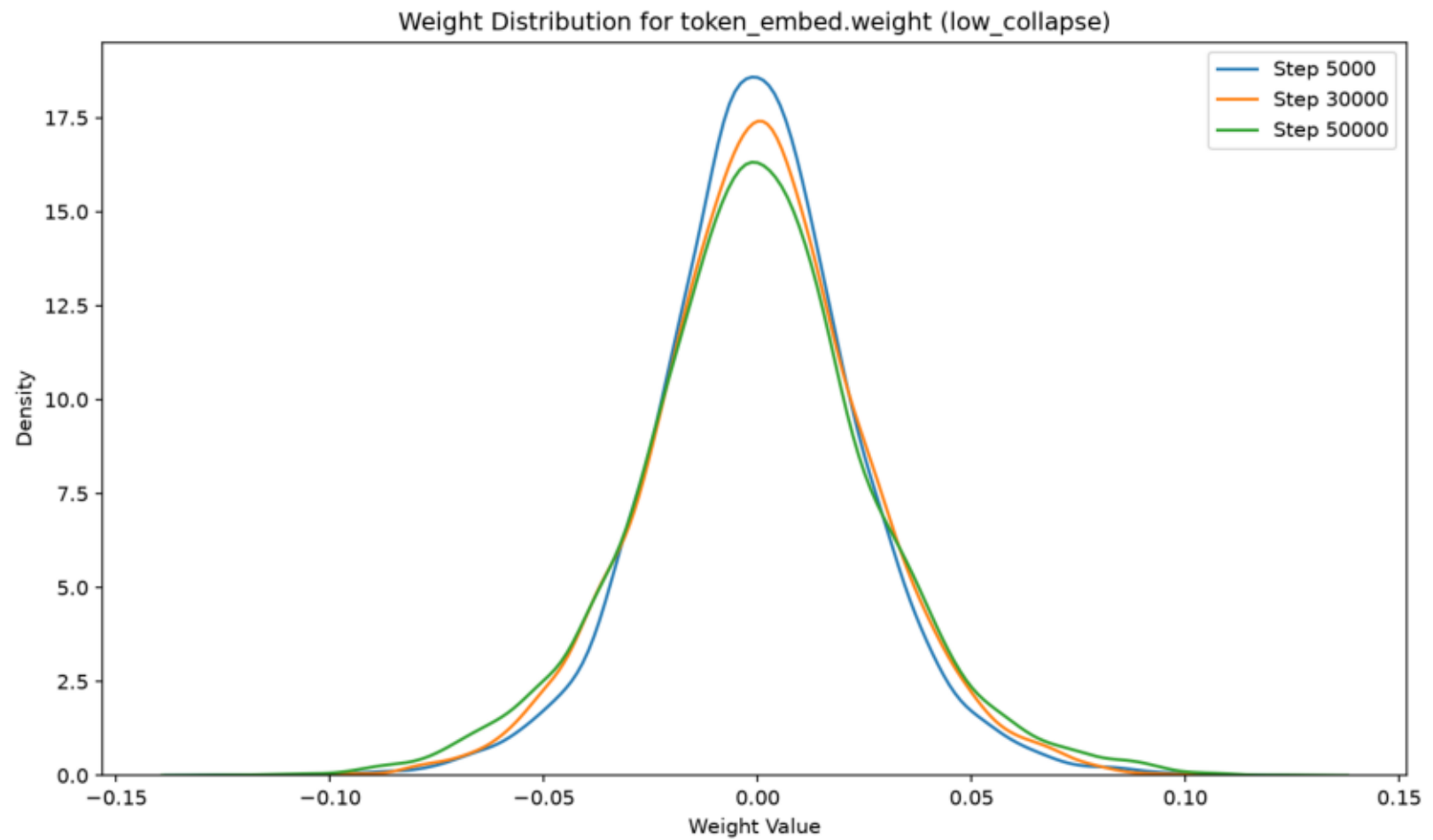
Singular Value Spectrum (severe)



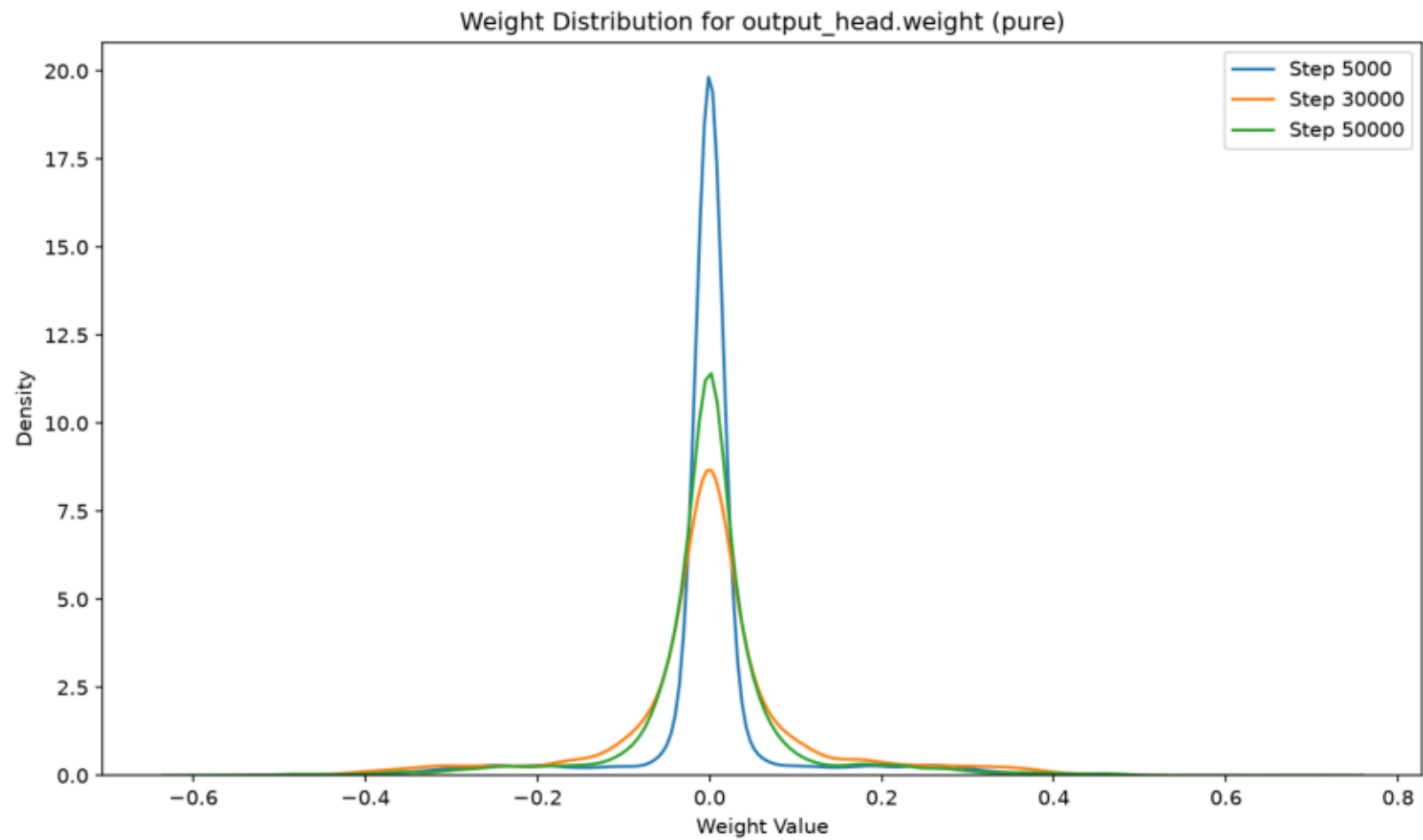
Weight Distribution: collapse (low)



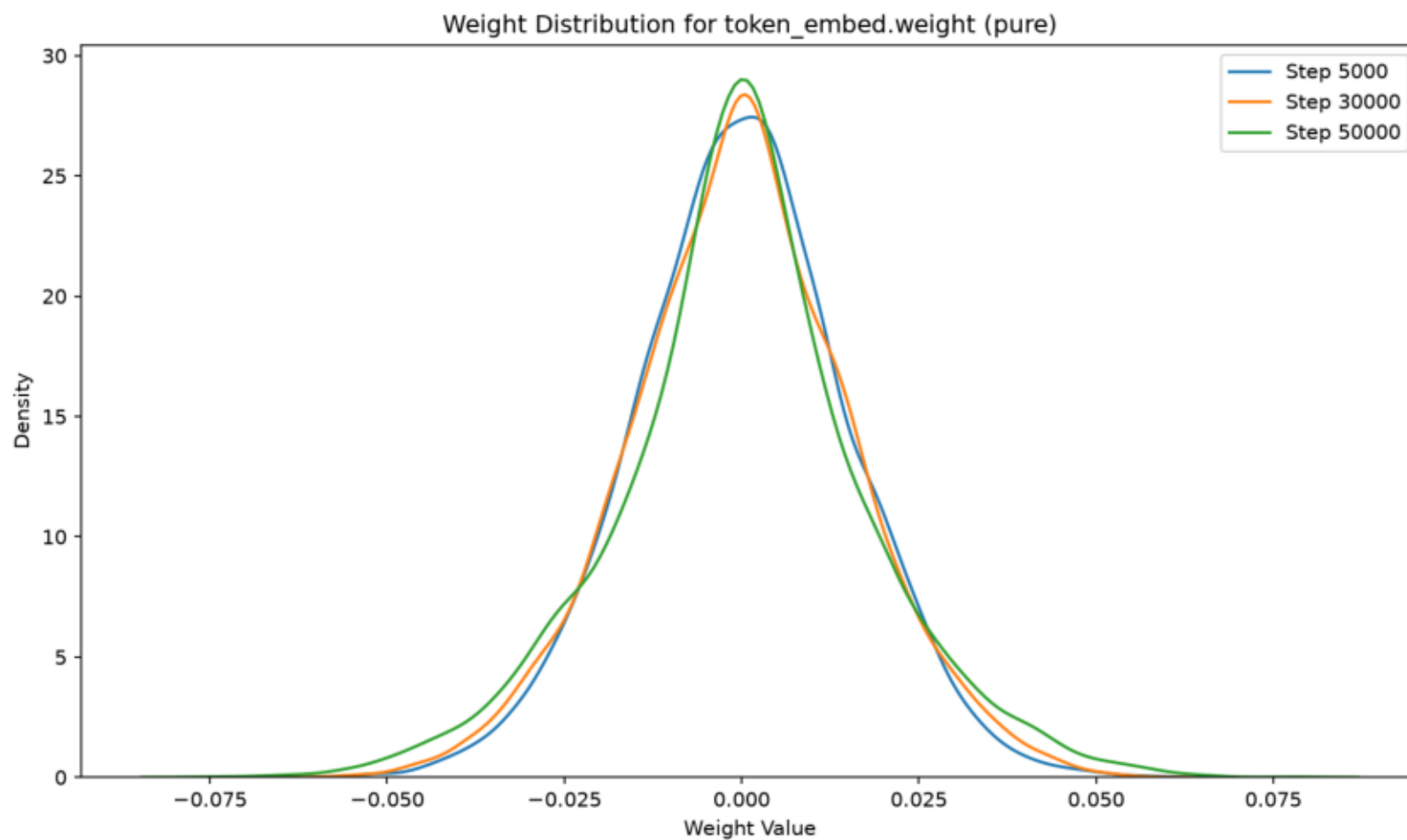
Weight Distribution: collapse (low)



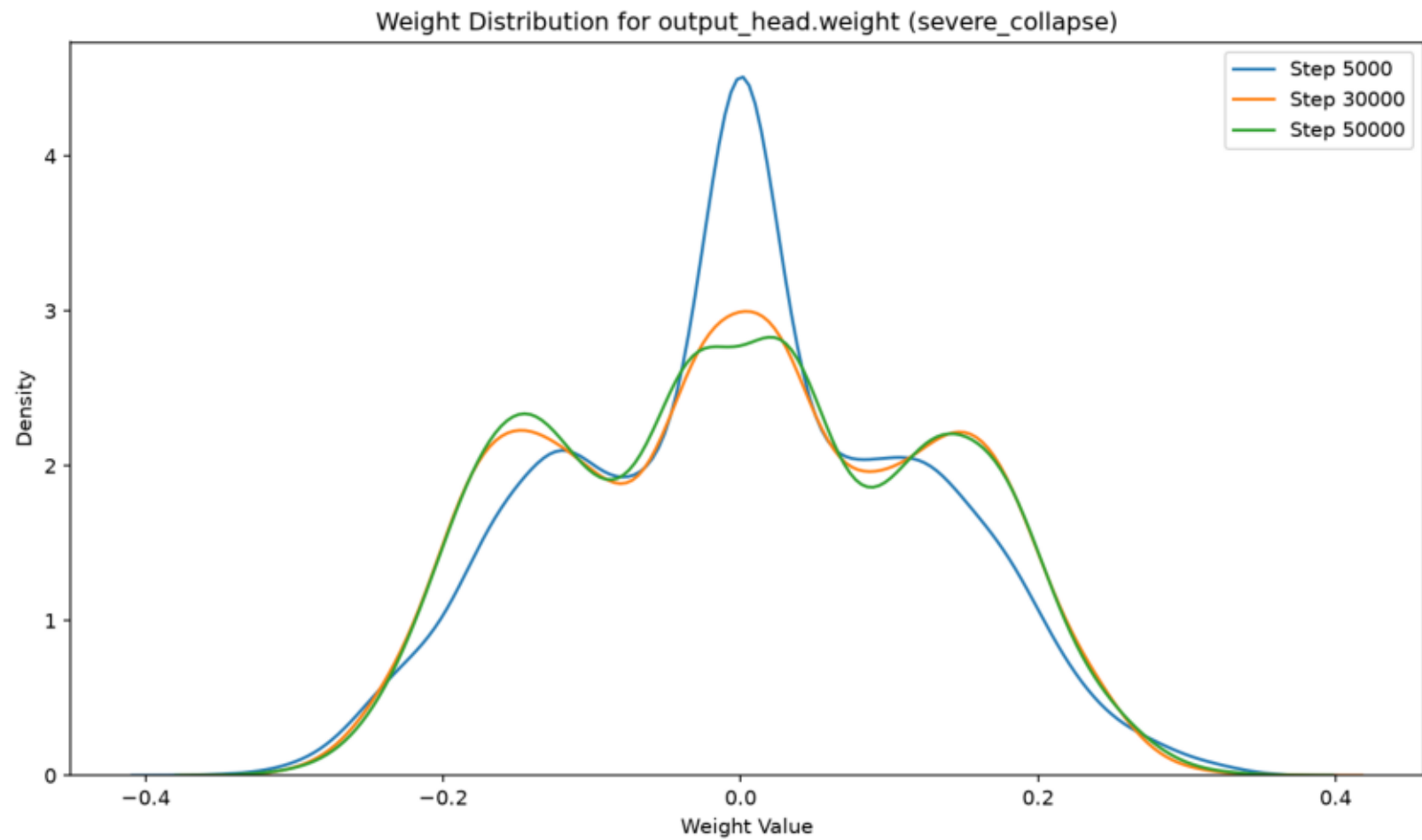
Weight Distribution: output (pure)



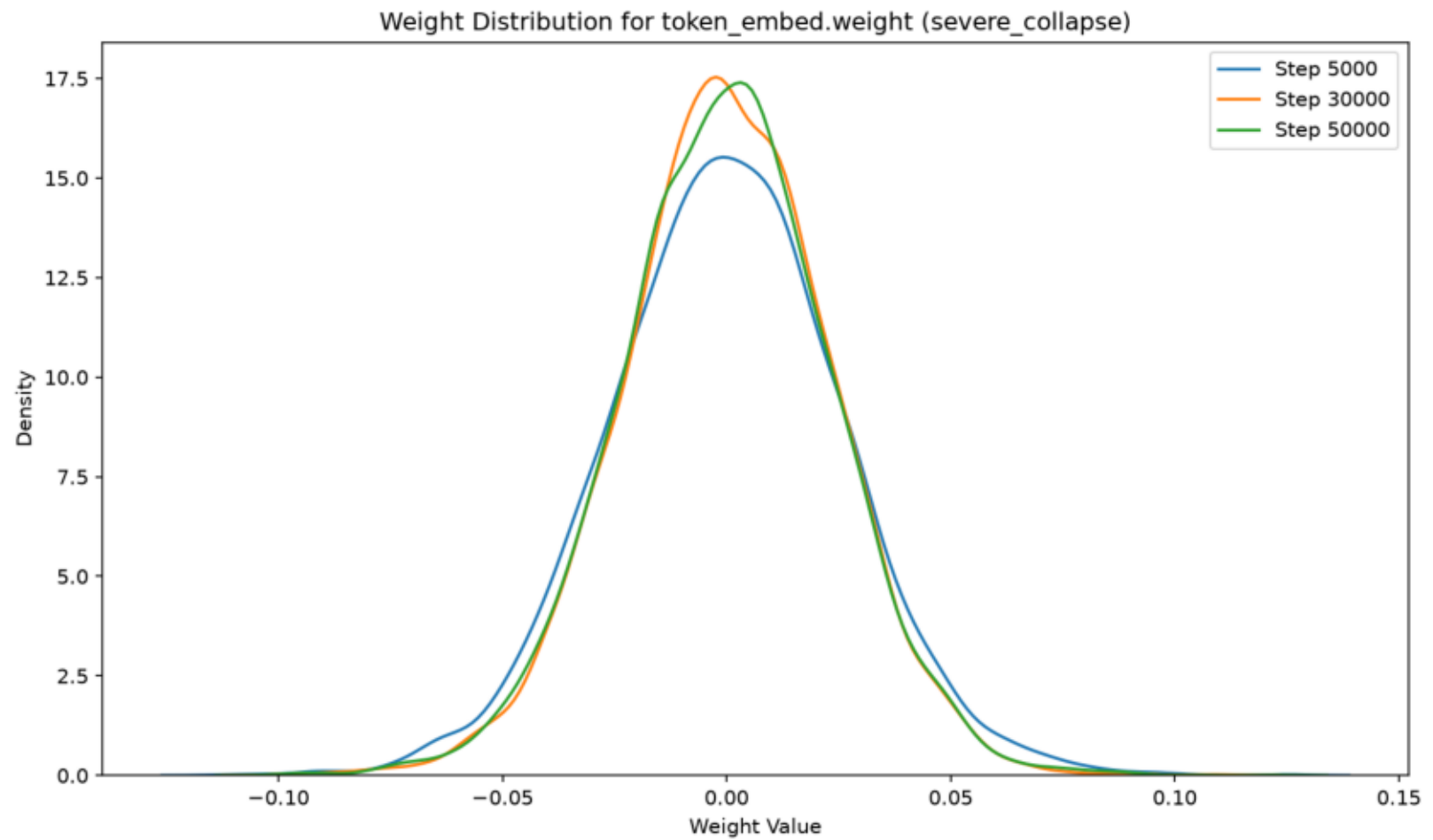
Weight Distribution: token (pure)



Weight Distribution: collapse (severe)

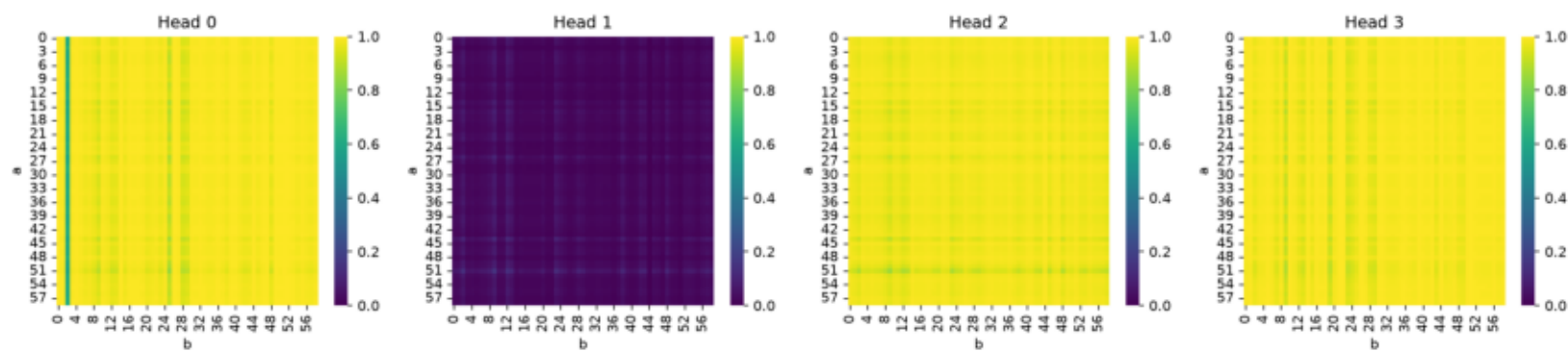


Weight Distribution: collapse (severe)



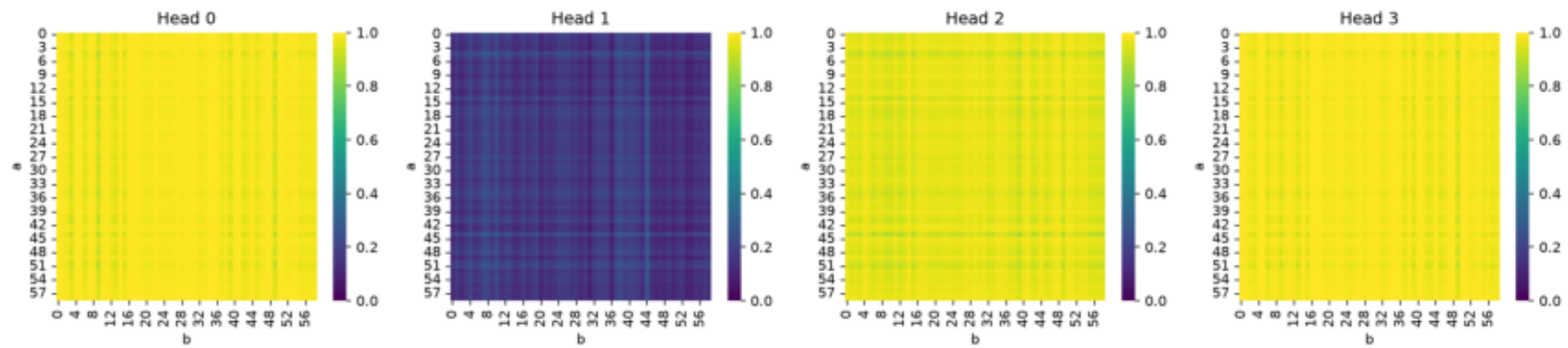
Attention Map: high Step 30000

Attention (pos 1 -> pos 0) at Step 30000 (high_collapse)



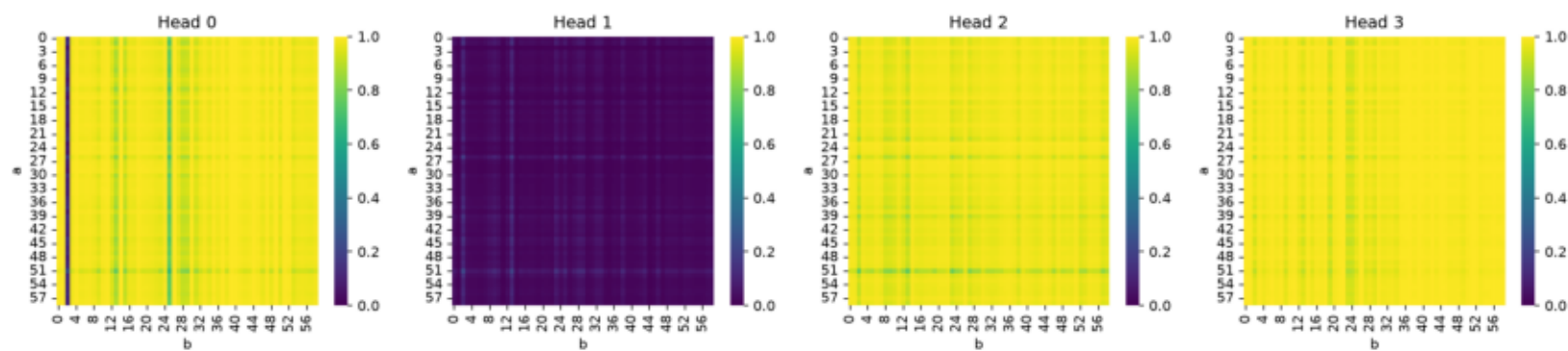
Attention Map: high Step 5000

Attention (pos 1 -> pos 0) at Step 5000 (high_collapse)



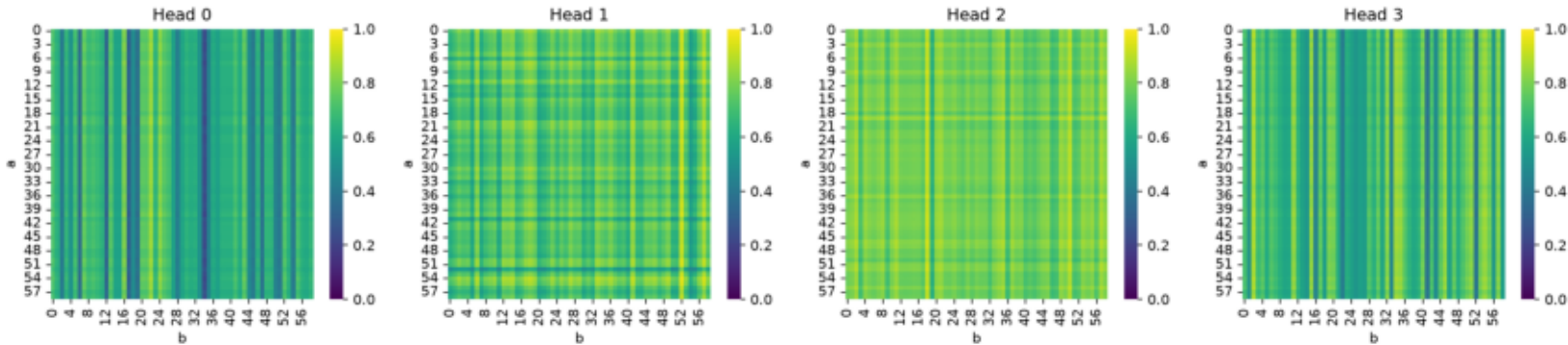
Attention Map: high Step 50000

Attention (pos 1 -> pos 0) at Step 50000 (high_collapse)



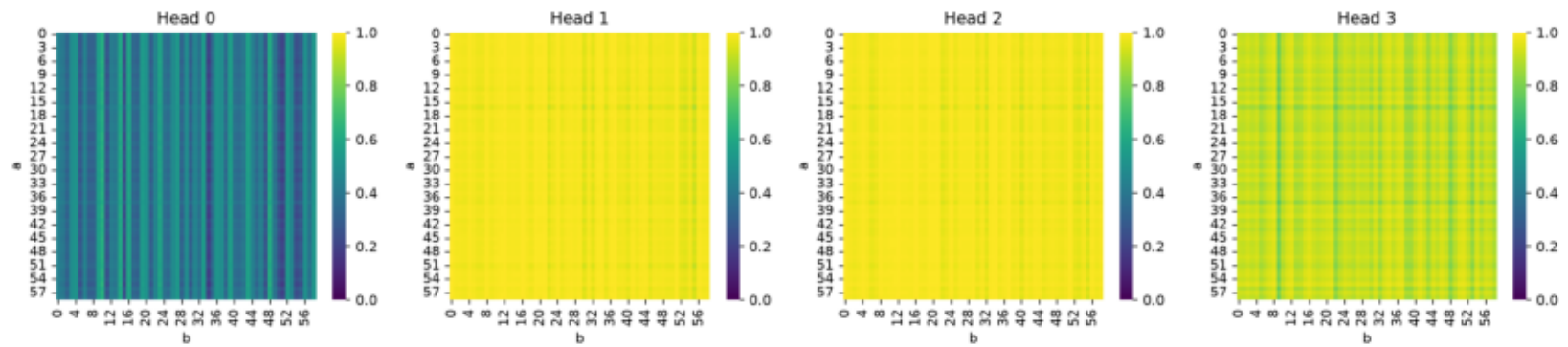
Attention Map: low Step 30000

Attention (pos 1 -> pos 0) at Step 30000 (low_collapse)



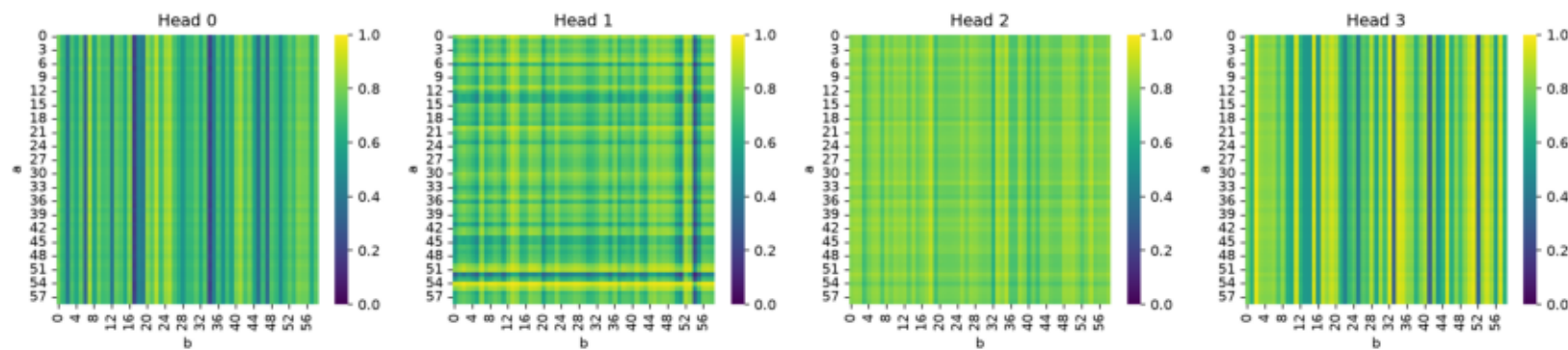
Attention Map: low Step 5000

Attention (pos 1 -> pos 0) at Step 5000 (low_collapse)



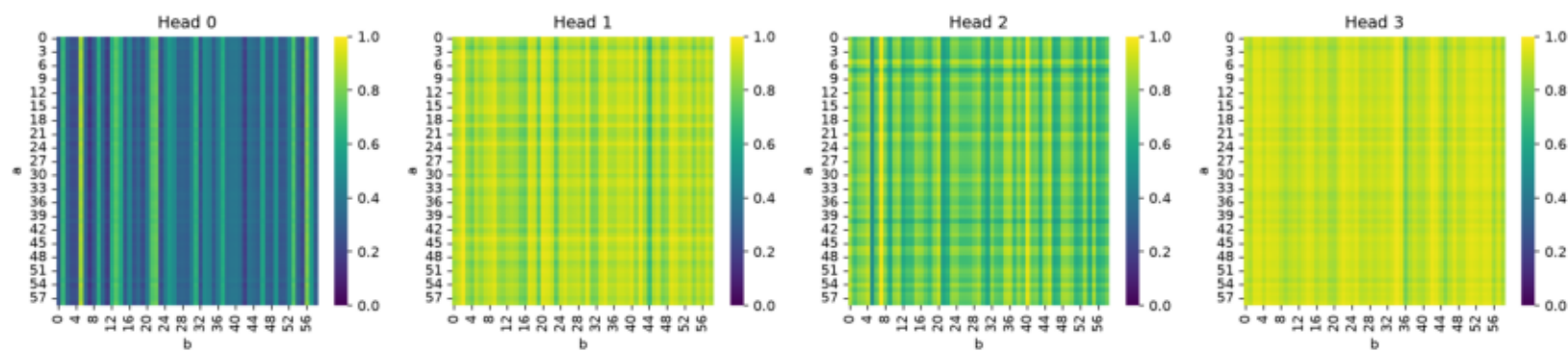
Attention Map: low Step 50000

Attention (pos 1 -> pos 0) at Step 50000 (low_collapse)



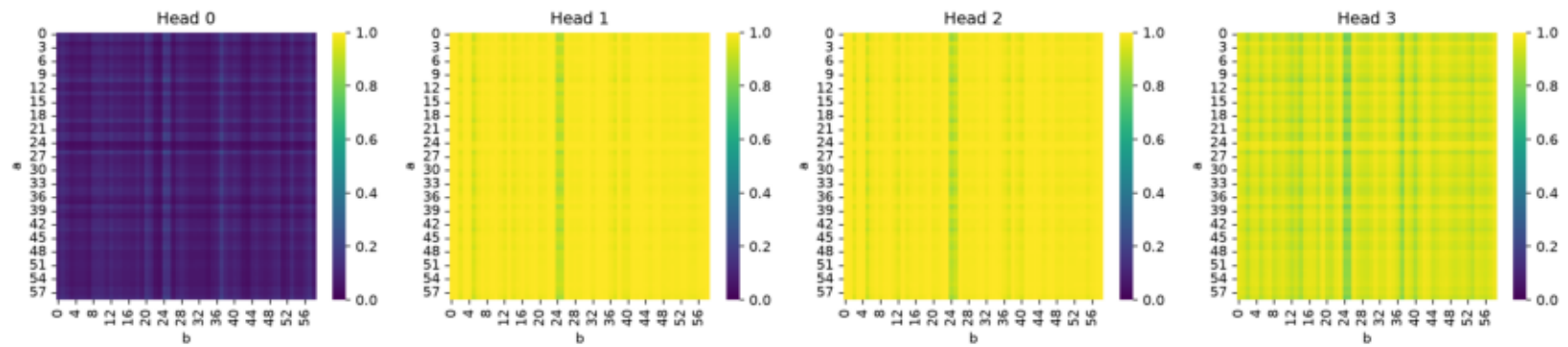
Attention Map: medium Step 30000

Attention (pos 1 -> pos 0) at Step 30000 (medium_collapse)



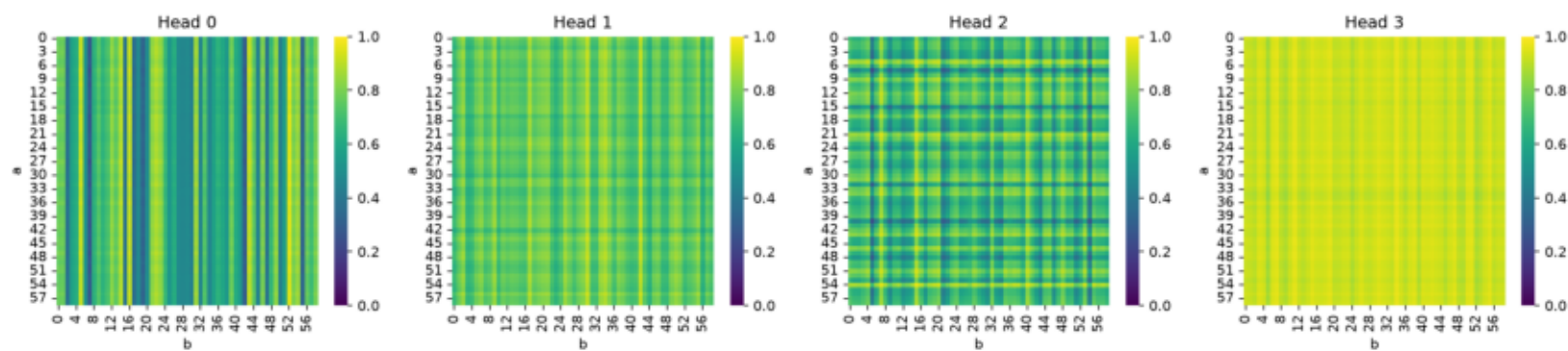
Attention Map: medium Step 5000

Attention (pos 1 -> pos 0) at Step 5000 (medium_collapse)



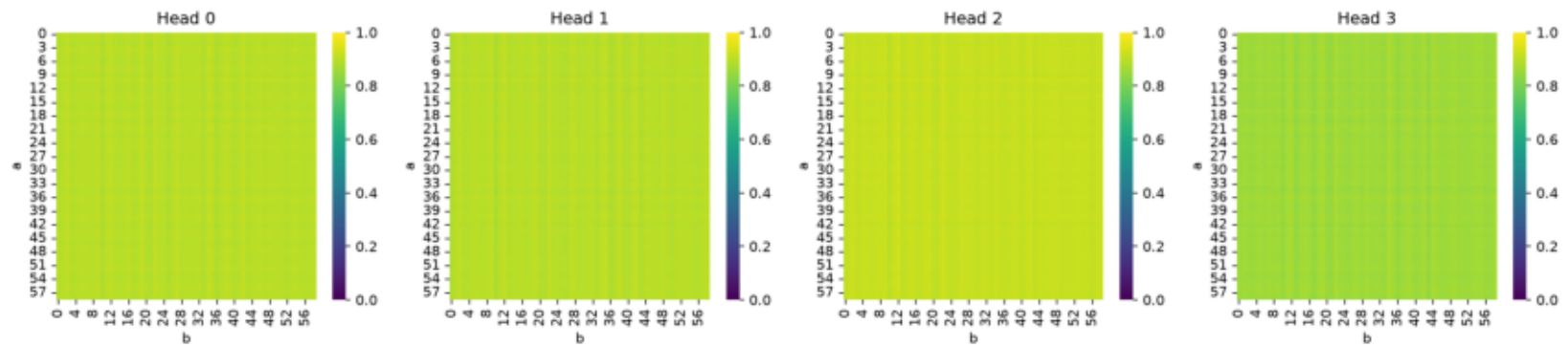
Attention Map: medium Step 50000

Attention (pos 1 -> pos 0) at Step 50000 (medium_collapse)



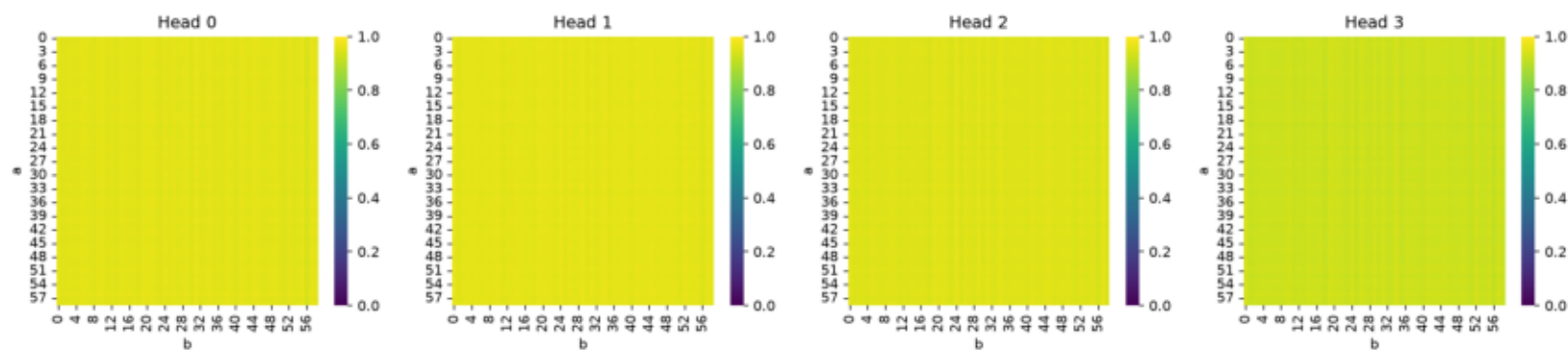
Attention Map: pure Step 30000

Attention (pos 1 -> pos 0) at Step 30000 (pure)



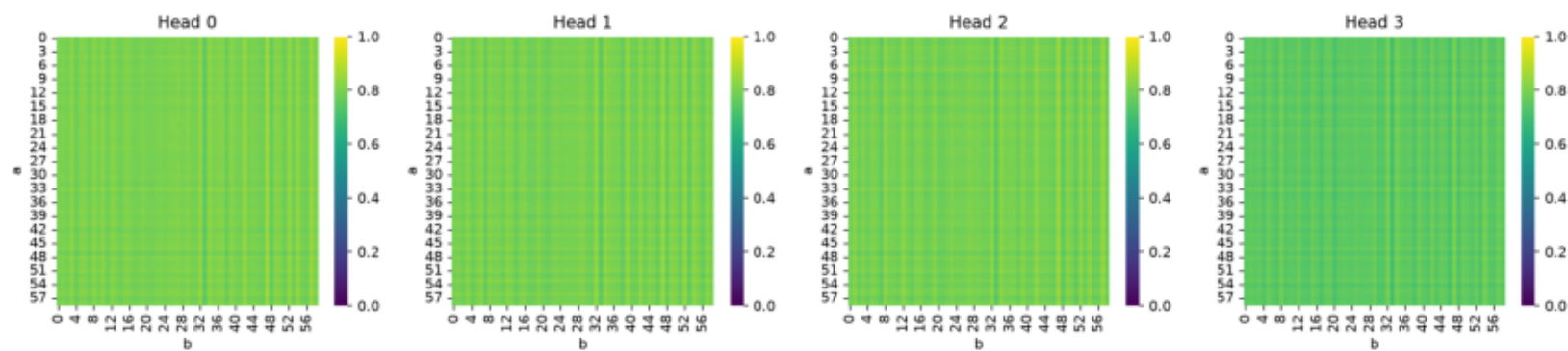
Attention Map: pure Step 5000

Attention (pos 1 -> pos 0) at Step 5000 (pure)



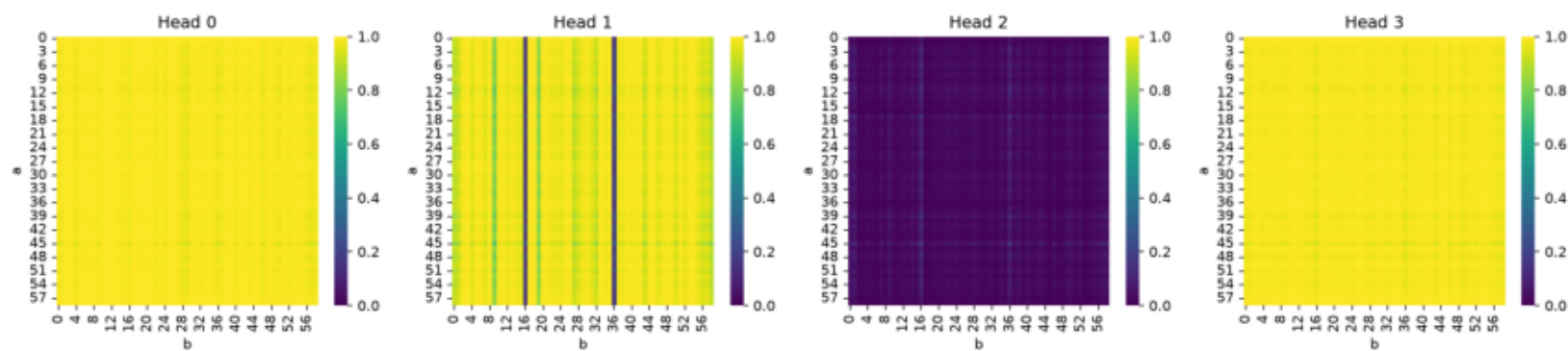
Attention Map: pure Step 50000

Attention (pos 1 -> pos 0) at Step 50000 (pure)



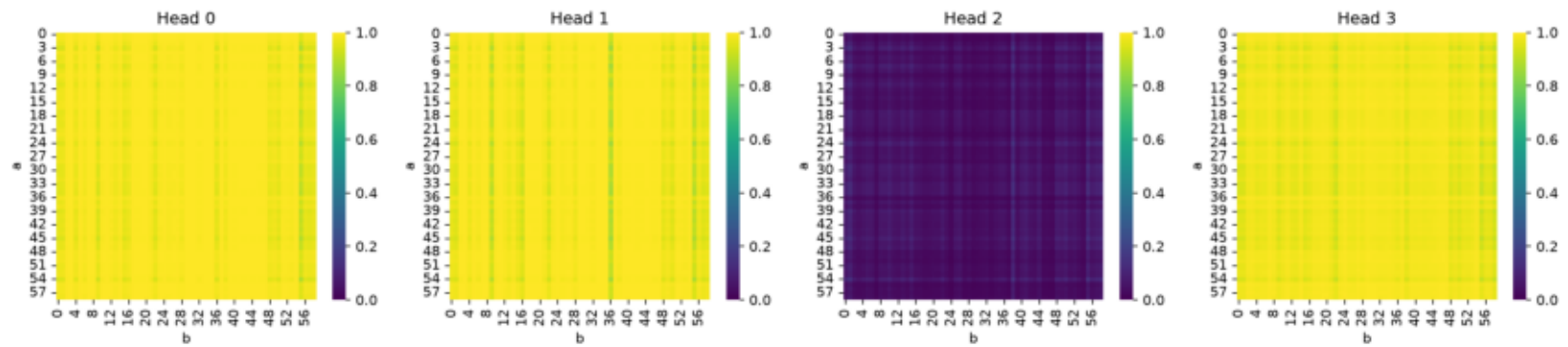
Attention Map: severe Step 30000

Attention (pos 1 -> pos 0) at Step 30000 (severe_collapse)



Attention Map: severe Step 5000

Attention (pos 1 -> pos 0) at Step 5000 (severe_collapse)



Attention Map: severe Step 50000

Attention (pos 1 -> pos 0) at Step 50000 (severe_collapse)

